

InventOR: A Human-Reviewed, Prompt-Configured Large Language Model Decision Support System for Material-Level Inventory Control

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Abstract

Material planners must set a reorder point, a safety stock and an order quantity for thousands of plant-material pairs, usually under ERP parameters that were last revised long ago. We define and evaluate *InventOR*, a decision support system (DSS) for this recurring decision. For each material, a large language model (LLM) configured only through a prompt, without training or fine-tuning, proposes a reorder-point/order-quantity (r, Q) policy with a written justification and an escalation flag. Deterministic software gates the proposal against minimum order quantities and storage limits and replays it on held-out demand before the planner, who remains the decision-maker, reviews it. We derive six design requirements and evaluate the artifact on 365 plant-material pairs from three plants of an industrial manufacturer, using three independent LLM runs, seven reference arms and an ablation that withholds the ERP planning parameters. In a hard-limit storage replay, the LLM policies cost about 11% less than ERP-derived parameters at a mean fill rate 0.1–0.4 points lower, but 6%, 8% and 18% more than

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history-only, Croston/SBA and empirical-quantile (r, Q) baselines. The saving comes from carrying 13–17% less inventory with 11–20% more stockout days; when shortages are priced at one unit cost, ERP-derived parameters are cheaper and a simulation-tuned policy is cheapest. Static storage checks are insufficient: simulated trajectories exceed storage limits for 53–66% of pairs. InventOR is a checkable, human-reviewed candidate generator, not an optimizer that beats data-driven baselines; benefits to planners are untested. We contribute design principles, review rules and a research agenda.

Keywords: Decision Support Systems, Large Language Models, Generative Artificial Intelligence, Inventory Control, Human–AI Decision Making, Design Science, Operations Research

1. Introduction

Replenishment parameters in manufacturing are set per plant-material pair: a reorder point, a safety stock and an order quantity. In large ERP systems these values often remain at levels set years earlier, while demand, supplier performance and storage constraints change. Revisiting them requires analysts who can fit demand and lead-time statistics, apply inventory theory, respect minimum order quantities (MOQs) and storage limits, and explain the result to planners. A conversational large language model (LLM) with a code interpreter appears to offer this scarce skill, and a growing literature explores such tools for supply-chain and operations decisions [1, 2, 3, 4, 5].

Three problems limit its use. First, LLM outputs can be unfaithful or fabricated [6], so a proposed policy is a candidate, not a decision. Second, superiority claims over ERP parameters are hard to interpret when the LLM and the comparators receive different information. Third, a storage rule that

15 holds when a policy is written down can be violated as stock builds up over
16 time.

17 This paper treats the problem as one of decision support system (DSS)
18 design. We define *InventOR* as a human-reviewed, model-driven DSS for
19 the recurring decision of setting replenishment parameters per plant-material
20 pair (Section 2). A planner remains the decision-maker. The system supplies,
21 for each material, a candidate (r, Q) policy proposed by an enterprise LLM
22 application that is configured only through a prompt and receives no project-
23 specific training or fine-tuning, together with a justification, an escalation
24 flag and a counterfactual simulation; deterministic software parses, gates
25 and simulates every proposal before the planner reviews it and alone decides
26 whether any ERP parameter is changed.

27 We follow a design-science logic [7, 8]: we state problem and scope, de-
28 rive design requirements, build the artifact and evaluate it on real industrial
29 data. The questions are how an LLM should be embedded so that its pro-
30 posals are bounded, verifiable, reproducible and reviewable; what cost and
31 service they achieve when the information available to each approach is con-
32 trolled; whether they respect storage limits along simulated trajectories and
33 are stable across runs; and when a planner should trust, escalate or reject a
34 proposal.

35 Our contributions are: (C1) a definition and scope of the DSS with six
36 design requirements that bound the role of the LLM (Section 2); (C2) an
37 architecture and operating rules that separate generation, deterministic gat-
38 ing, simulation and planner review, with design principles (Sections 4 and 8);
39 (C3) an information-controlled, trajectory-level evaluation with seven refer-
40 ence arms, an ablation denied the ERP planning parameters and a hard-
41 limit storage replay (Section 5); and (C4) complete-cohort evidence from

three LLM runs on 365 plant-material pairs, with managerial and research implications.

Our main findings are as follows. Against ERP-derived parameters the LLM policies are about 11% cheaper at a mean fill rate 0.1–0.4 points lower, but the LLM costs 6%, 8% and 18% more than history-only, Croston/SBA and empirical-quantile (r, Q) baselines at essentially equal service, and its saving over ERP comes from carrying less inventory at the price of more stockout days. Storage limits that hold in a static check are violated along unconstrained trajectories for 53–66% of pairs. The simulation therefore does not show superior optimization by the LLM; it shows that a prompt-configured LLM can produce checkable candidate policies without code or training, whose benefit to planners remains to be tested.

2. Problem, Scope and Definition

2.1. The decision problem

The decision supported is the periodic (re)setting of a reorder point, a safety stock and an order quantity for each plant-material pair held in an ERP system. In the classification of Gorry and Scott Morton [9] it is a partly structured operational-control decision: the calculation is textbook, but the choice of estimator, the treatment of intermittent or trending demand, the plausibility of recorded lead times and the handling of constraints require judgement that is applied inconsistently when thousands of materials compete for a few analysts' time. The decision-maker is the material planner; hard constraints are the minimum order quantity (MOQ) and a physical storage limit; and feedback is slow, because a changed parameter shows its effect only in later demand. The scope is one pair at a time (single item, single

67 echelon, continuous review, lost sales, (r, Q) family); order release, supplier
68 selection, forecasting-model development and multi-echelon networks are out
69 of scope, and the LLM is a candidate generator that does not forecast, place
70 orders, simulate or decide.

71 2.2. Definition and level of automation

72 **Definition.** *InventOR is a human-reviewed, model-driven deci-*
73 *sion support system that, for one plant-material pair at a time, (i)*
74 *obtains from a large language model, configured only by a written*
75 *prompt, a candidate (r, Q) policy with a justification and an es-*
76 *calation flag; (ii) verifies the candidate deterministically against*
77 *explicit constraints; (iii) evaluates it by counterfactual simulation*
78 *against reference policies on held-out demand; and (iv) presents*
79 *the result to a planner, who alone decides whether ERP param-*
80 *eters are changed.*

81 In the classical DSS architecture of data, model and dialog components
82 [10, 11], the LLM and the simulator form the model component, the material-
83 scoped data extract and the prompt form the data and knowledge compo-
84 nents, and the review interface is the dialog component (Section 4.1). In
85 terms of the levels of automation of Parasuraman et al. [12], InventOR op-
86 erates at a low-to-intermediate level: it selects and suggests one alternative
87 and leaves execution to the human. Accepting automated suggestions invites
88 complacency and automation bias [13], which the review rules of Section 4.3
89 are meant to counter.

90 2.3. Design requirements

91 The requirements follow from the known unreliability of LLM outputs
 92 [6], guidance on human–AI interaction and accountability [14, 15], run-to-
 93 run nondeterminism of LLM inference [16] and the constraints of inventory
 94 control. Table 1 ties each to a design feature (Section 4) and an evaluation
 95 (Section 6).

Table 1: Design requirements, the design features that address them and where they are evaluated.

ID	Requirement	Design feature	Evaluation
DR1	Bounded generation: the LLM proposes and has no authority	Narrow candidate-generator role; prohibited methods; no order placement	Sections 6.1, 6.3
DR2	Verifiable feasibility along time, not only at a point	Deterministic gate; hard-limit trajectory replay	Sections 6.1, 6.5
DR3	Counterfactual evidence for the reviewer	Held-out simulation against reference arms	Sections 6.2, 6.8
DR4	Information transparency: comparisons control what each arm sees	Seven reference arms; LLM ablation without ERP parameters	Sections 6.3, 6.4
DR5	Reproducibility and accountability of the configuration	Frozen prompt file; per-call log; repeated runs	Section 6.7
DR6	Human control with explicit escalation	Trust, escalate and reject rules; escalation flag	Sections 4.3, 6.6

96 3. Related Work and Positioning

97 *Decision support systems and human–AI decision making.* DSS research
 98 has long distinguished data-, model- and knowledge-driven systems and has
 99 stressed that the value of a DSS lies in improving the judgement of a hu-
 100 man decision-maker rather than in replacing it [9, 10, 11]. Research on
 101 human–automation interaction shows that the way suggestions are scoped,
 102 presented and verified determines whether users rely on them appropriately
 103 or over-rely on them [12, 13],. Recent DSS work derives design principles
 104 for LLM-based assistants [17], reports design and pilot implementations of
 105 AI-enabled decision support in operations [18], and shows that the uptake of

106 algorithmic advice depends on transparency and on the user’s metacognition
107 [19, 20]; evaluations of data-driven forecasting systems in practice likewise
108 stress lessons beyond accuracy [21]. Model-driven support also shapes how
109 groups handle disagreement [22], and human override of algorithmic safety-
110 stock advice is a documented risk [23]; InventOR therefore keeps the planner
111 as decision-maker and makes the evidence for each proposal explicit.

112 *LLMs in operations and supply-chain decisions..* Reviews and frameworks
113 map the opportunities for generative AI in supply chains [1, 2, 24, 25, 3, 4, 26]
114 and stress that adoption depends on assigned human roles, data quality and
115 governance [5, 27, 28, 29, 30]. Evidence from manufacturing planning is thin:
116 a fine-tuned LLM for digital-manufacturing optimization [31], a transformer
117 trained to set order timing and quantity and benchmarked against (r, Q) [32],
118 and experiments showing that ChatGPT reproduces human decision biases
119 of managers [33]. Most technical LLM work translates problem statements
120 into optimization models and solver code [34, 35, 36, 37], studies multi-agent
121 control in inventory flows [38, 39], disruption modelling [40] or classical OR
122 tasks such as routing [41]. The closest LLM study to ours uses an LLM
123 as a forecasting copilot for spare-parts demand, combined with simulation,
124 and reports high variability for irregular demand that can be reduced by
125 averaging repeated queries and enriching prompts [42]. We test run-to-run
126 variability directly (Section 6.7); the difference is that our LLM proposes a
127 complete policy, not a forecast.

128 *Data-driven inventory control without LLMs..* Learning-based replen-
129 ishment is well established, including data-driven policies learned from
130 sequentially observed non-stationary demand [43] and forecasts augmented
131 with leading indicators [44], and heuristics remain competitive in constrained

132 settings such as perishable inventory [45]. Deep reinforcement learning has
 133 been applied to omnichannel and multi-echelon systems [46, 47, 48] and
 134 aerospace replenishment [49]; deep-learning forecasting has been integrated
 135 with replenishment [50]; contextual distributionally robust optimization
 136 uses side information to set order quantities [51]; and distribution-free
 137 online algorithms handle capacitated warehouses under uncertain price and
 138 demand [52]. The closest operational neighbour is a trained data-driven
 139 framework that jointly sets replenishment parameters, safety stock and
 140 safety lead time on real automotive data [53]. Reinforcement learning has
 141 also been combined with demand forecasting to optimize order points [54].
 142 These methods need training or tuning; InventOR needs none, but its
 143 generator cannot be inspected beyond its written justification.

144 *Intermittent and spare-parts demand..* Material histories in manufacturing
 145 are frequently intermittent or lumpy. The statistical literature offers special-
 146 ized estimators and classification schemes [55, 56, 57, 58, 59], and recent stud-
 147 ies develop feature-based forecast combinations and temporal aggregation for
 148 intermittent demand [60, 61], forecasting for spare parts across product life-
 149 cycle phases [62], and integrated forecasting and planning or multi-echelon
 150 stocking models for repairable and spare parts [63, 64].

151 InventOR is not a new inventory model and does not claim to beat
 152 trained or optimized methods; its (r, Q) policy is the standard workhorse
 153 of continuous-review control [65, 66, 67].

154 4. The InventOR Decision Support System

155 4.1. Architecture and components

156 Following the classical DSS components [10, 11], InventOR has a *data*
157 component (a material-scoped extract of daily demand, lead times, working-
158 day flags, cost and constraint parameters and ERP reference values; the
159 fields supplied are controlled and logged; DR4), a *knowledge* component (the
160 system prompt file defining the task, the prohibited methods, the storage
161 rules and the output schema, Supplementary File S1; DR1, DR5), a *model*
162 component split into a generator (an enterprise LLM application with a code
163 interpreter, configured only by the prompt, proposing one (r, Q) policy with
164 a justification and an escalation flag; DR1) and a deterministic verifier and
165 simulator (parser, gate, held-out simulation with hard storage-limit replay
166 and reference arms; DR2, DR3), a *dialog* component (policy, justification,
167 flag and simulated comparison shown to the planner under the trust, escalate
168 and reject rules; DR6) and a *governance* layer (frozen prompt file, per-call
169 log of input hash, time, session and raw response, repeated runs; DR5). The
170 generator is the only non-inspectable component. In the evaluated prototype
171 the dialog component is the per-pair output record; no interactive interface
172 was built or evaluated.

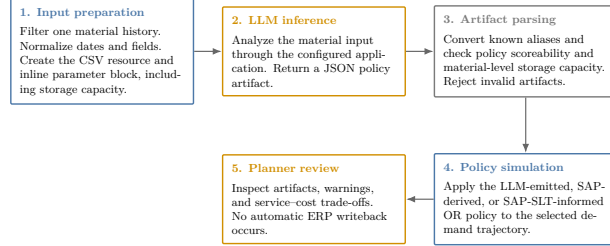
173 4.2. Operation in four steps

174 For one material, InventOR answers how much safety stock to hold, at
175 what level to reorder and how much to order, in four steps (Figure 1).

176 **Step 1. Generate.** The LLM, accessed through an enterprise application
177 with a code interpreter, receives one material’s history (daily de-
178 mand, realized lead times, working-day flags) and a short parameter

InventOR prototype workflow

Executed material-level path from prepared history to planner-reviewable simulation output.



Deterministic components reproduce simulation results from stored artifacts; LLM inference itself may vary.

Figure 1: InventOR workflow: material data are prepared, an LLM proposes a policy, deterministic software parses, checks and simulates it, and a planner reviews the result.

block (unit cost, holding rate, ordering cost, MOQ, service target, storage limit, and current ERP safety stock and safety lead time for comparison). A written prompt (the only configuration; no weights are changed) [68] instructs it to compute demand and lead-time statistics, apply an economic-order-quantity (EOQ) argument and a normal-approximation safety stock, respect the storage limit, and report a structured policy with a short justification and escalation flag.

Step 2. Gate. Deterministic code extracts the policy, rejects malformed or infeasible outputs (missing or non-numeric fields, non-positive reorder point or order quantity, negative safety stock, reorder point below safety stock, safety stock plus order quantity above the storage limit) and retries the generation up to three times. The backtest applies one further admission check, that the order quantity is not below the MOQ.

Step 3. Simulate. Each accepted policy is replayed on the material’s demand observed *after* the data cutoff, a held-out evaluation in the

196 sense of out-of-sample forecast testing [69, 70], together with the
197 comparator policies, producing holding and ordering cost, fill rate
198 and stockout days.

199 **Step 4. Review.** A planner sees the policy, the LLM’s justification and
200 flag, and the simulated comparison. Only the planner changes ERP
201 parameters. The planner’s decision (accept, adjust or reject), the
202 review time and an optional free-text comment are logged with
203 the call, so that practitioner feedback accumulates as a by-product
204 of use; the prototype evaluated here does not yet implement this
205 logging.

206 The LLM is a *candidate generator* whose output is never trusted without
207 the gate; it does not forecast, place orders or run the simulation. The prompt
208 prohibits Monte Carlo simulation, distribution fitting, machine-learning li-
209 braries and specialized intermittent-demand estimators, so that the policy
210 rests on transparent statistics.

211 4.3. *Trust, review and reject rules*

212 We propose three rules for planners, in line with guidelines for human–AI
213 interaction and AI auditing [14, 15]. The rules and the 0.5-point threshold
214 are design proposals not evaluated with planners; the trust rule would be
215 met by a minority of pairs (the LLM is cheaper than the history-only OR in
216 74–77 of 346).

217 **Trust (apply after review)** when the proposal passes the gate, the sim-
218 ulated trajectory respects the storage limit when orders are truncated at it,
219 and the simulated cost is not above that of the best history-only OR ref-

220 erence (Croston/SBA or empirical quantile) with a fill rate not more than
221 about 0.5 points lower.

222 **Escalate** when the LLM raises an escalation flag, the material is lumpy or
223 has strongly seasonal or trending demand, the plant or calendar data were
224 repaired, or the policy differs strongly between repeated generations.

225 **Reject** when the proposal fails the gate after the retry budget, exceeds the
226 storage limit in the hard-limit replay, or is dominated (higher cost and lower
227 fill) by a history-only reference.

228 *4.4. Provenance and reproducibility of the configuration*

229 The system prompt, a plain-text file deployed unchanged in the enter-
230 prise application, defines the task, forbidden methods, capacity rules and
231 output schema (Supplementary File S1). Every call logs the SHA-256 of its
232 input, UTC time, session identifier and raw response. The platform does
233 not expose the model identifier or sampling settings; the evaluated model
234 (Anthropic Claude Sonnet 4.6 with a code interpreter) is operator-declared.
235 Reproducibility rests on the prompt file, the call logs and three independent
236 runs (Section 6.7).

237 **5. Application and Evaluation Design**

238 *5.1. Evaluation approach*

239 The evaluation addresses the design requirements (Table 1). The pri-
240 mary basis is the hard-limit replay with working-day receipts and no short-
241 age penalty; sensitivity analyses vary these choices. Following design science
242 [7, 8], the evaluation is artificial and ex post in the framework of Venable

et al. [71]: real data and a faithful simulation but no live users. It establishes technical efficacy, feasibility and reliability, not usability, adoption or improved planner decisions.

5.2. Industrial application and data

The application is a pilot at a large manufacturer. The dataset contains 233,078 daily records for 411 plant-material pairs from three plants (A, B and C), with a data cutoff on 10 March 2026 and a post-cutoff horizon observed to 27 August 2026. Of the 411 pairs, 365 are eligible for scoring (234 in Plant A, 65 in Plant B, 66 in Plant C) and are sent to the LLM. For 19 pairs (all in Plant A) the MOQ plus the reference safety stock exceeds the storage limit, so no comparator can satisfy both rules; these are excluded from outcome comparison, leaving a *common-feasible cohort* of 346 pairs (215, 65 and 66). Demand is concentrated (the five largest materials account for 40.3% of demand). The source working-day flag of Plant C is all zero; we reconstruct it from the same-date majority of Plants A and B and test this repair in Section 6.8. Data quality is a known threat to decision support [72] and inventory records [73, 74, 75].

Each pair is sent as a material-scoped CSV with daily demand, recorded lead times, a working-day flag, the minimum order quantity, unit cost, holding-cost rate, ordering cost, service-level target, the storage limit, and two *ERP planning parameters* (current safety stock and safety lead time), which are withheld in the ablation arm. The CSV also carries supplier and lane context fields that the prompt does not use.

5.3. Reference arms and information content

ERP safety stock and safety lead time embed planner judgement and are alternative ways of buffering unreliability [76], and planned lead times can

drift upward [77]. The LLM and the classical comparators therefore do not necessarily use the same information, and we construct arms that make it explicit (Table 2).

Table 2: Comparison arms and the information each uses. All arms respect MOQ and are projected to the storage limit when possible.

Arm	Parameters from	ERP used	inputs	Purpose
SAP-derived	ERP safety stock; EOQ order quantity	Safety stock		ERP-derived reference (not observed practice)
SLT-informed OR	Stationary (r, Q) with ERP safety lead time added to the lead time	Safety lead time		OR policy using the ERP safety lead time
ERP-floor OR	SLT-informed OR with safety stock floored at the ERP value	Both		OR arm matched to the LLM's full information
History-only OR	Stationary (r, Q) from demand and lead-time history, mean lead time	None		ERP-independent baseline
Croston/SBA OR	(r, Q) with the Syntetos–Boylan-approximated Croston demand rate and its forecast-error spread, mean lead time	None		Intermittent-demand baseline
Empirical-quantile OR	(r, Q) with the reorder point set to the service-level quantile of historical rolling lead-time demand	None		Distribution-free baseline
Simulation-tuned (r, Q)	Reserve factor and order-quantity multiple chosen by simulation on the last 250 working days before the cutoff	None		Trained data-driven baseline
LLM (runs 1–3)	LLM proposal with all CSV fields (Section 5.2)	Both		Main treatment
LLM without ERP	LLM with ERP safety stock and safety lead time set to “not supplied”	None		Ablation: what the LLM does when it cannot anchor on ERP

Three further arms use no ERP input and represent stronger data-driven practice. The *Croston/SBA OR* replaces the mean daily demand by the Croston estimate with the Syntetos–Boylan bias correction [55, 56] (smoothing constant 0.1), uses the root mean squared one-step forecast error as demand spread and the history-only lead-time moments, which suits inter-

277 mittent demand [57]. The *empirical-quantile OR* sets the reorder point to the
 278 service-level quantile of historical rolling lead-time demand over the rounded
 279 mean lead time. The *simulation-tuned* (r, Q) policy selects, from a grid of
 280 reserve factors and order-quantity multiples around the history-only policy,
 281 the cheapest candidate whose fill rate on the last 250 working days *before*
 282 *the cutoff* meets the service target (or the best-fill candidate if none does).
 283 No post-cutoff data are used in any of these arms. We do not implement a
 284 deep reinforcement-learning baseline [47, 46]; this is a limitation (Section 8).

285 5.4. *Simulation, capacity replay and metrics*

286 Each pair is simulated independently on the post-cutoff horizon as a
 287 single-item, single-echelon continuous-review (r, Q) system [65, 78, 79, 67, 66]
 288 with lost sales [80], in the spirit of simulation-based experiments on supply
 289 and timing uncertainty [81], a shared opening stock, an empty pipeline and
 290 arrivals after the rounded mean lead time in working days (Appendix Ap-
 291 pendix A). MOQs are treated as hard order-size constraints, as imposed
 292 by suppliers [82], and storage limits as capacity bounds on the replenish-
 293 ment decision [52]. The primary metrics are the mean material-horizon
 294 holding-plus-ordering cost, the mean material-level fill rate, the aggregate
 295 (volume-weighted) fill rate and the number of stockout days. We report
 296 paired differences with 10,000-sample bootstrap 95% intervals.

297 A static check $SS + Q \leq \text{max_storage_units}$ does not show that storage
 298 limits hold as inventory evolves, because the reorder point includes lead-time
 299 demand and inventory on order can exceed the limit on arrival. We therefore
 300 record the post-receipt on-hand stock for every policy and day and report
 301 the share of pairs and days above the limit (the *unconstrained* simulation),
 302 and we *replay* every policy with each order truncated to the storage limit

303 minus the current inventory position (the *hard-limit replay*), which is our
304 primary evaluation.

305 5.5. Repeated runs and sensitivities

306 To examine run-to-run variation, the entire cohort is generated three
307 times (runs 1, 2 and 3) with the same prompt file and the same data. Run 1
308 serves as the reference for tabulated results unless stated otherwise; runs 2
309 and 3 provide replicates. Sensitivity analyses vary the working-day recon-
310 struction (majority, any-plant, strict-all, exclusion of Plant C and calendar-
311 day arrivals), the shortage penalty (zero, one and five times unit cost).

312 6. Results

313 6.1. Completion and policy characteristics

314 All four runs (three with the full input and one ablation, Section 6.4)
315 return a valid, storage-feasible (r, Q) policy for all 365 pairs after at most
316 three attempts per pair. The validity gate rejects non-positive reorder points
317 and order quantities, negative safety stocks and reorder points below the
318 safety stock. The ERP safety stock is zero for 198 of the 365 pairs (188 of
319 the 346 in the cohort below), so for a majority of pairs the SAP-derived arm
320 holds no safety stock at all.

321 6.2. Headline comparison with ERP-derived parameters

322 Unless stated otherwise, all results use the hard-limit replay on the
323 346-pair common-feasible cohort with working-day receipts and no shortage
324 penalty (Section 6.5 gives the unconstrained simulation). Table 3 compares
325 all arms; its last columns give the LLM cost relative to each arm in the three
326 runs (positive: the LLM costs more). In run 1 the LLM policies cost 11.4%

less than SAP-derived policies and 30.5% less than the SLT-informed OR (mean 2,928 versus 3,305 and 4,211), at a mean fill 0.18 points lower than SAP-derived (98.11% versus 98.30%; interval -0.54 to $+0.10$). The mean cost difference against SAP-derived is -377 (95% interval -651 to -154); the LLM is cheaper in 241 of 346 pairs and, in 226, simultaneously no more expensive and no worse in fill. It incurs more stockout days (910 versus 804).

Relative to SAP-derived parameters the LLM carries about 17% less average inventory (16,256 versus 19,565 units) and about 16% lower holding cost at about the same number of orders per year (17.3 versus 17.6), and it accepts 11–20% more stockout days (runs 3 and 2). The saving is therefore a trade of inventory for service, not a better ordering rhythm, and Section 6.8 shows that once shortages carry a cost this trade no longer favours the LLM.

Table 3: Hard-limit replay, common-feasible cohort ($n = 346$), working-day receipts, zero shortage penalty. Cost is the mean material-horizon holding-plus-ordering cost; average inventory is in units; orders are per year. The last column gives the LLM cost relative to the arm in runs 1/2/3 (positive: the LLM costs more).

Arm	Mean cost	Holding	Ordering	Avg. inv.	Orders	Mean fill (%)	Aggr. fill (%)	Stockout days	LLM vs arm (%)
ERP-derived (SAP)	3,305	2,172	1,132	19,565	17.6	98.30	79.34	804	$-11.4 / -11.3 / -11.4$
SLT-informed OR	4,211	2,608	1,603	21,614	24.5	98.23	79.15	831	$-30.5 / -30.4 / -30.5$
ERP-floor OR	4,248	2,633	1,616	21,937	24.7	98.23	79.15	831	$-31.1 / -31.0 / -31.1$
History-only OR	2,764	1,725	1,039	15,972	16.3	98.14	78.72	874	$+5.9 / +6.0 / +5.9$
Croston/SBA OR	2,705	1,665	1,040	15,521	16.3	98.30	78.87	803	$+8.2 / +8.4 / +8.3$
Empirical-quantile OR	2,489	1,543	946	13,331	14.8	98.35	78.87	747	$+17.7 / +17.8 / +17.7$
Simulation-tuned (r, Q)	3,266	2,473	793	11,914	12.6	96.05	84.84	1,973	$-10.4 / -10.3 / -10.3^\dagger$
LLM (run 1)	2,928	1,824	1,104	16,256	17.3	98.11	78.87	910	
LLM (run 2)	2,931	1,850	1,081	16,599	17.0	97.89	78.70	968	
LLM (run 3)	2,929	1,841	1,088	16,958	17.0	98.16	78.90	890	
LLM without ERP	3,014	1,902	1,112	16,507	17.3	98.06	78.89	890	$-8.8^\ddagger$ (vs SAP)

[†] Other comparisons of the ablation are in Section 6.4. [‡] Different operating point (mean fill 96.05%, aggregate fill 84.8%); the paired interval of the cost difference includes zero, and with priced shortages this policy is the cheapest arm (Section 6.8).

6.3. Information asymmetry: what the LLM adds beyond the ERP inputs

First, the SLT-informed OR and its ERP-floor variant are the most expensive arms (4,211 and 4,248): the ERP safety lead time inflates their effective lead time and buffers. Against them the LLM is 30–31% cheaper in every run, which mostly measures how heavily ERP inputs over-buffer.

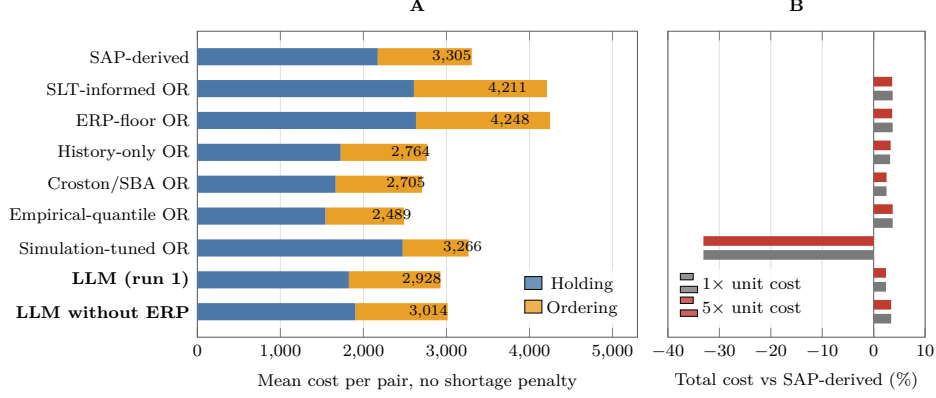


Figure 2: Cost structure and effect of pricing shortages (hard-limit replay, 346 pairs; LLM rows in bold). (A) Mean cost per pair split into holding and ordering cost, with the total at the end of each bar. The LLM and the data-driven ORs carry less inventory than SAP-derived parameters and the ERP-informed ORs. (B) Total cost when lost sales are charged at one and five times the unit cost, relative to SAP-derived parameters (negative: cheaper). Once shortages are priced SAP-derived parameters beat the LLM, and the simulation-tuned policy is cheapest.

Second, the *history-only OR*, which uses no ERP inputs, costs 2,764 at a fill of 98.14%. The LLM costs 5.9–6.0% more in all three runs (run 1: +164, interval +91 to +240), with fill differences of -0.03 , -0.25 and $+0.01$ points whose intervals include zero, and is cheaper in only 74–77 of 346 pairs. The stronger data-driven arms widen the gap: the LLM costs 8.2–8.4% more than the Croston/SBA OR (2,705, fill 98.30%; run 1: +223, interval +97 to +362; cheaper in 91–99 pairs) and 17.7–17.8% more than the empirical-quantile OR (2,489, fill 98.35%; +439, interval +329 to +557; cheaper in 44–49 pairs; mean fill 0.24 points lower). Part of the LLM’s advantage over ERP-derived parameters therefore reflects weak ERP benchmarks.

The simulation-tuned (r, Q) policy sits at a different operating point: an aggregate fill of 84.8% and the lowest inventory (11,914 units) but a mean fill of 96.05% and 1,973 stockout days, because it is tuned on the pre-cutoff year. The LLM costs 10.3–10.4% less at zero penalty with a 2.1-point higher

mean fill, but the interval includes zero ($-2,527$ to $+857$); once shortages are priced the tuned policy is by far the cheapest arm (Section 6.8).

Third, the LLM improves on ERP-derived parameters (11.3–11.4% cheaper at 0.14–0.40 points lower fill) but not on the best classical baselines, which also yield a policy for every material at lower cost.

6.4. LLM without ERP planning parameters (ablation)

In this ablation the LLM receives the same inputs as in runs 1–3, but the ERP safety stock and safety lead time are blanked in the CSV and marked “not supplied”; the system prompt file is unchanged. It tests whether the LLM anchors on the ERP planning parameters. All 365 pairs returned a valid, storage-compliant policy, and the policies change only slightly beyond run-to-run variation: median ratios to run 1 are 1.00, and the reorder point differs by more than 10% in 54.6% of pairs (48.3% and 48.0% between ERP-supplied runs). In the hard-limit replay the cost is 3,014 at a mean fill of 98.06%, against 2,928–2,931 in runs 1–3. It is 8.8% below SAP-derived parameters (interval -566 to -61 ; fill -0.24 points, interval including zero) and 28.4–29.1% below the SLT-informed and ERP-floor ORs, but 9.0% above the history-only OR (interval $+134$ to $+387$; cheaper in 91 of 346 pairs), 11.4% above the Croston/SBA OR, 21.1% above the empirical-quantile OR and 7.7% below the simulation-tuned policy. By plant the pattern is unchanged (-10.8% in Plant A, -21.9% in Plant B, $+41.6\%$ in Plant C against $+36.2\%$ in run 1), so the Plant C loss does not stem from reliance on ERP parameters. The information-asymmetry concern therefore does not explain the LLM’s advantage over ERP-derived parameters, and the ERP planning fields appear to be of limited use to the LLM. The ablation withholds only these two fields; unit cost, holding rate, ordering cost, MOQ, service target and

384 storage limit remain available, as they are for all reference arms.

385 6.5. Storage limits along simulated trajectories

386 All 365 LLM policies satisfy the static rule $SS+Q \leq \text{max_storage_units}$,
387 yet in the unconstrained simulation post-receipt on-hand stock exceeds the
388 limit for 183 pairs (52.9%; 178–183 across runs; 10.2% of pair-days) for the
389 LLM, 198 (57.2%) for SAP-derived and 230 (66.5%) for the SLT-informed
390 OR, because inventory on order and lead-time demand add to stock above
391 $SS + Q$. The static check is thus necessary but not sufficient for any
392 method. Under the hard-limit replay no arm exceeds the limit by more
393 than one unit; enforcing it raises the LLM mean cost from 2,850 to 2,928
394 and SAP-derived from 3,184 to 3,305 without changing the ranking.

395 6.6. Plants and demand regimes

396 The average hides a strong plant effect (Figure 3, hard-limit replay,
397 run 1). The LLM costs 14.1% less than SAP-derived in Plant A (cheaper in
398 179 of 215 pairs) and 22.7% less in Plant B (54 of 65), but 36.2% *more* in
399 Plant C (cheaper in 8 of 66); the Plant C loss is similar in the other full runs
400 (+32.2%, +31.5%) and persists under every working-day reconstruction in
401 which Plant C can be scored (Section 6.8). Against the data-driven base-
402 lines the LLM costs more in every plant: 7.2%, 3.6% and 6.7% more than the
403 history-only OR, 13.1%, 2.3% and 6.2% more than the Croston/SBA OR,
404 and 20.8%, 7.7% and 29.5% more than the empirical-quantile OR in Plants
405 A, B and C. The SLT-informed and history-only ORs coincide in Plant C
406 because the ERP safety lead time is zero for all its pairs.

407 By demand regime, the 33 lumpy materials show the largest reduction
408 against SAP-derived (−25.4% in run 1) and the 313 smooth materials −9.8%

409 at a 0.24-point lower mean fill. On lumpy materials the service effect is
 410 inconsistent (mean fill +0.38 points in run 1, -0.29 in run 2), the history-only
 411 and Croston/SBA ORs cost about 9–10% less than the LLM, and only the
 412 empirical-quantile OR is slightly more expensive (1.2%); with 33 materials
 413 this is indicative only. By terciles of the coefficient of variation, the LLM
 414 costs 15.7%, 7.5% and 12.9% less than SAP-derived (lowest to highest), with
 415 a 0.68-point lower mean fill in the middle tercile.

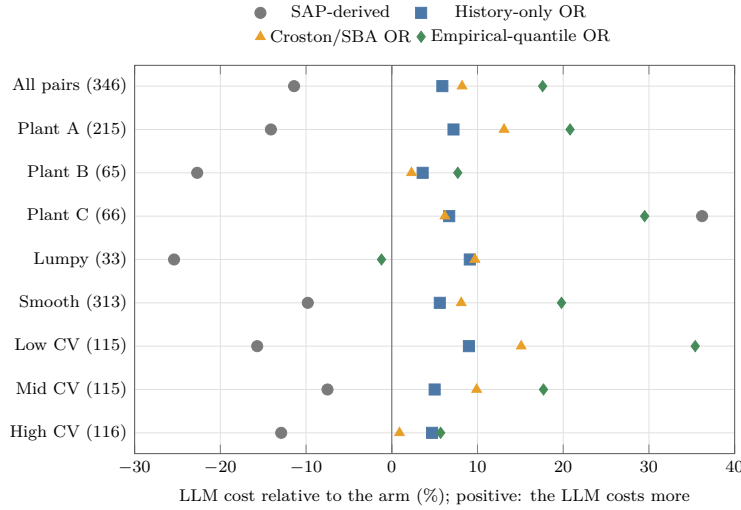


Figure 3: Where the LLM wins and loses (run 1, hard-limit replay). Each marker is the LLM’s mean cost relative to one comparator in a segment of pairs (number of pairs in brackets); markers to the right of the vertical line mean the LLM costs more. The LLM is cheaper than SAP-derived parameters in Plants A and B but not in Plant C, and costs more than the three data-driven baselines in almost every segment.

416 6.7. Run-to-run stability

417 Across the three runs the aggregate results are very similar: the LLM
 418 cost relative to SAP-derived is -11.4% , -11.3% and -11.4% , and the policy-
 419 value medians are close (reorder point 1,584, 1,582 and 1,466; order quantity
 420 2,737, 2,777 and 2,693; safety stock 683, 688 and 669). Individual policies

421 vary, however. Between runs 2 and 3 on the 365 pairs, all three values are
 422 identical for only 36 pairs; every policy resolves to the (r, Q) family, and the
 423 median absolute differences are 81 units (reorder point), 6 (order quantity)
 424 and 3 (safety stock). The reorder point differs by more than 10% in about
 425 48% of pairs, and a few pairs vary strongly (one reorder point was 14,882,
 426 14,947 and 29,218 units). Aggregate behaviour is reproducible, but a single
 427 generation per material is a noisy estimate that should be reviewed [16].

428 *6.8. Sensitivity analyses*

429 *Plant C calendar..* Plant C’s working-day flag required reconstruction, so we
 430 re-scored all policies under four alternatives (any-plant, strict-all, exclusion
 431 of Plant C, calendar-day arrivals). Where Plant C can be scored, the LLM
 432 remains 31–35% more expensive than SAP-derived there in the three full
 433 runs, and the cohort-wide advantage is -10.8% to -11.1% ; the strict and
 434 exclusion variants (280 pairs) show -17.0% to -17.6% . The Plant C result
 435 is therefore not an artefact of the repair, although the LLM saw demand
 436 series built on the repaired calendar.

437 *Shortage penalty..* Figure 2B gives mean total cost per pair (holding, order-
 438 ing and shortage cost) relative to SAP-derived parameters in the hard-limit
 439 replay for penalties of one and five times the unit cost. Costs are dominated
 440 by the shortage term, and the differences among the non-tuned arms (0.1–
 441 3.7%) should be read with caution. The simulation-tuned policy is by far
 442 the cheapest (1,094,043 at one times the unit cost, about 33% below SAP-
 443 derived), because it was selected to meet the service target on the pre-cutoff
 444 history. Among the other arms SAP-derived parameters are the cheapest,
 445 as they carry more inventory and fewer stockouts. LLM run 1 then costs
 446 2.4% more than SAP-derived at both multiples, is about as expensive as the

447 Croston/SBA OR (1,673,003 versus 1,674,925 at one times), and is cheaper
 448 than the SLT-informed (1.2%), ERP-floor, empirical-quantile and history-
 449 only (0.8%) ORs. The LLM’s cost advantage over ERP-derived parameters
 450 is therefore specific to the case in which service shortfalls are not charged.

451 7. Managerial Insights and Best Practices

452 1. **Position the tool as a reviewed decision support system, not**
 453 **an autonomous engine.** Single generations differ at the material level
 454 (Section 6.7). The gate-simulate-review structure yields a justified can-
 455 didate for every material and leaves the decision to the planner; whether
 456 this helps planners is untested.

457 2. **Benchmark against data-driven baselines.** ERP-derived parameters
 458 and ERP-informed ORs are weak benchmarks (safety stocks are often
 459 stale or zero; safety lead times inflate buffers), so comparing only with
 460 ERP overestimates the benefit: the LLM cost 6%, 8% and 18% more than
 461 history-only, Croston/SBA and empirical-quantile baselines.

462 3. **Expect plant-level differences.** Against ERP-derived parameters the
 463 LLM was cheaper in Plants A and B (14–23%) and on 33 lumpy materials
 464 (about 25%, mixed service) but not in Plant C; we did not test whether
 465 escalation rules would catch such cases.

466 4. **Verify storage along the trajectory.** The static rule $SS + Q \leq \text{limit}$
 467 cannot protect a warehouse: 53–66% of pairs breach the limit in simula-
 468 tion under each of the three arms checked. Enforce a cap in the order-
 469 release logic or treat the limit as a design requirement.

- 470 **5. Price shortages explicitly.** Without a shortage penalty the ranking
 471 rewards under-stocking; with a penalty of one unit cost the LLM is about
 472 2.4% more expensive than SAP-derived. Set service targets and penalties
 473 before reading cost comparisons.
- 474 **6. Freeze and log the configuration.** Treat the system prompt as a
 475 controlled file, log every call (input hash, time, session, raw response)
 476 and re-run before changing the prompt.
- 477 **7. Scale in stages.** Automate generation and gating first, improve data
 478 quality (calendars, lead-time records, stale ERP parameters) next, refine
 479 the prompt under freeze-and-log, and widen the scope last; re-run the
 480 evaluation after every change.
- 481 **8. Cost and latency.** Each generation took several minutes, whereas OR
 482 baselines take milliseconds; periodic regeneration would support adapt-
 483 able planning [83].

484 **8. Discussion, Limitations and Research Perspectives**

485 *What the evidence shows..* A prompt-configured LLM generates feasible, re-
 486 viewable (r, Q) policies for every material; in simulation they cost less than
 487 ERP-derived parameters but more than data-driven OR policies. InventOR
 488 is therefore not presented as outperforming optimization: its role is a re-
 489 viewed candidate generator, and whether that helps planners is a hypothesis
 490 the simulation cannot test. Against the ERP-informed arms alone the LLM
 491 would look far better (-11.4% against SAP-derived, -30.5% against the
 492 SLT-informed OR).

493 *Design principles for LLM-based decision support.* We derive six principles
 494 from this application and offer them as hypotheses for design-science work
 495 [7]. **DP1, bound the generator:** the LLM is a candidate generator whose
 496 prompt fixes the method, constraints and output schema, with no authority
 497 to act. **DP2, verify deterministically and along time:** check outputs
 498 with code and feasibility along the trajectory, because point checks can pass
 499 while constraints are violated over time (Section 6.5). **DP3, show counter-**
 500 **factual evidence:** give the reviewer simulated comparisons with credible
 501 references. **DP4, control information in evaluation:** compare against
 502 baselines with and without the same information and withhold incumbent
 503 inputs in an ablation (Sections 6.3, 6.4). **DP5, freeze and log:** treat the
 504 prompt as a controlled file, log every call and repeat runs (Section 6.7). **DP6,**
 505 **escalate by rule:** define in advance when proposals are trusted, escalated
 506 or rejected, to limit complacency [13] (Section 4.3).

507 *Implementation experience and roadmap.* The planner responsible for the
 508 pilot materials reported being satisfied with the tool. This single, unstruc-
 509 tured report is not evidence of benefit and is not part of the evaluation. The
 510 planned expansion from validated concept to scalable capability has four
 511 stages, each changing the configuration and therefore requiring the evalua-
 512 tion protocol to be repeated: (1) *automation*, connecting the generator to a
 513 workflow-automation tool and the enterprise agent platform so that gener-
 514 ation, gating and simulation run without manual uploads (planned for the
 515 fourth quarter of 2026 to the first quarter of 2027); (2) *data optimization*,
 516 improving data quality, such as the calendars and stale ERP parameters that
 517 mattered in Plant C, and integrating further relevant data; (3) *prompt opti-*
 518 *mization*, refining the prompt for more consistent outputs (Section 6.7); and

519 (4) *scaling and integration* across use cases and into the surrounding
520 landscape.

521 *Limitations..* (i) The data come from one manufacturer and three plants.
522 (ii) The model identifier and inference settings are operator-declared. (iii)
523 No planner outcomes were measured (informal planner feedback aside): the
524 benefits of reviewability, auditability and planner support and the rules of
525 Section 4.3 are design intentions that a field study must test. (iv) The sim-
526 ulation uses deterministic arrival at the rounded mean lead time, ignores
527 observed receipts, corrections and scrap, starts with an empty pipeline and
528 does not simulate lead-time variability, which may bias comparisons that
529 depend on safety lead time; we did not report service-target attainment. (v)
530 The primary results use a zero shortage penalty; with a positive penalty
531 the ranking changes. Costs are dominated by a few high-demand materials
532 and the 346 pairs are not independent, so the bootstrap intervals under-
533 state uncertainty. (vi) The (r, Q) family is single-item and single-echelon.
534 (vii) Individual policies vary across runs, and Plant C required calendar re-
535 construction. (viii) We included no deep reinforcement-learning or neural
536 forecasting baseline, and the ERP-derived arm (ERP safety stock with our
537 own order quantity) is not an observed record of practice.

538 *Research perspectives..*

539 1. *Ensembling and hybrids.* Aggregating several generations may reduce
540 individual-policy noise [42], and combining the best data-driven OR policy
541 with LLM review or regime-specific choice [84] could capture the OR cost
542 level and the LLM’s flags.

543 2. *Comparison with trained methods.* Benchmarks against deep

reinforcement-learning and other trained policies [53, 46, 47] on identical data and information, and richer policy families such as (s, S) [58, 59], would clarify the trade-off between setup effort and policy quality.

3. *Planner-in-the-loop field trials.* The decision log of the review step provides the data for such studies. Prospective studies should compare planners with and without the aid on matched materials, measuring planner time, override rates and realized service and inventory, and should examine adoption drivers [28, 27, 5].

9. Conclusion

We defined InventOR as a human-reviewed decision support system for a recurring production-planning decision, the setting of replenishment parameters per plant-material pair, and evaluated it against six design requirements on a three-plant industrial dataset. A prompt-configured LLM generated feasible, justified (r, Q) policies for every material without training, and a deterministic gate, a hard-limit simulation and explicit review rules made them checkable. The proposals cost about 11% less than ERP-derived parameters at slightly lower service, but the LLM costs about 6%, 8% and 18% more than history-only, Croston/SBA and empirical-quantile policies; when lost sales are priced at one unit cost or more, ERP-derived parameters are the cheapest of the non-tuned arms and a simulation-tuned policy is cheapest overall. Storage limits that hold in a static check are exceeded along simulated trajectories for every method, and the LLM loses to ERP at one plant. The evidence supports InventOR as a checkable candidate generator inside a reviewed process, not as an optimizer that beats data-driven baselines; the

design principles of Section 8 are hypotheses, and field evidence on planner behaviour and realized outcomes is the central open step.

Appendix A. Formal Definitions

Reference calculation. These are the standard textbook calculations [67, 78]. For pair i , let $\bar{d}_i$ and $\sigma_{D,i}$ be the mean and standard deviation of pre-cutoff working-day demand, L_i and $\sigma_{L,i}$ the mean and standard deviation of the lead time in working days, K_i the ordering cost, h_i the annual holding cost per unit (unit cost times holding rate), z_i the standard normal quantile of the service target, and 250 the working days per year. The order quantity is $Q_i = \max\{\text{MOQ}_i, \lceil \sqrt{2K_i \cdot 250\bar{d}_i/h_i} \rceil\}$. The lead-time-demand standard deviation is $\sigma_{LTD,i} = \sqrt{L_i\sigma_{D,i}^2 + \bar{d}_i^2\sigma_{L,i}^2}$, the safety stock $SS_i = \lceil z_i\sigma_{LTD,i} \rceil$ and the reorder point $R_i = \lceil \bar{d}_iL_i^{\text{eff}} + SS_i \rceil$, where L_i^{eff} is the mean lead time for the history-only OR and the mean lead time plus the ERP safety lead time for the SLT-informed OR and the ERP-floor OR. The SAP-derived arm uses the ERP safety stock and the same Q_i . All policies are projected to $SS_i + Q_i \leq \text{max_storage_units}_i$ when $\text{MOQ}_i + SS_i$ fits.

Simulation. On each post-cutoff working day, demand is served from on-hand stock, unmet demand is lost, and an order of size Q_i is placed when the inventory position (on-hand plus on order) is at or below R_i and arrives after the rounded mean lead time in subsequent working observations. Observed receipts, corrections and scrap are ignored. The total cost is $TC_i = h_i\bar{I}_iT/250 + K_iN_i$, where $\bar{I}_i$ is the mean on-hand stock over the T simulated working days and N_i the number of orders. Fill rate is served divided by demanded units. In the hard-limit replay an order is truncated to $\text{max_storage_units}_i$ minus the inventory position, and no order is placed

594 if the truncated quantity is zero.

595 **Author Contributions**

596 Conceptualization, Methodology, Software, Formal analysis, Investiga-
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603 The author declares no known competing financial interests or personal
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605 paper.

606 **Data and Code Availability**

607 The raw operational data and material-level artifacts cannot be publicly
608 released because they contain commercially sensitive material, supplier
609 and planning information. A confidentiality-screened release package
610 containing analysis code, the anonymized variable schema, aggregate
611 evaluation results, manuscript sources and prompt files is available at
612 github.com/DressPD/inventor_public. A versioned Zenodo archive of
613 the release is available at <https://doi.org/10.5281/zenodo.22262052>.
614 Supplementary File S1 contains the system prompt used for all runs; the
615 prompt file, the runner code and, for each call, a hash of the input, the

616 UTC time, the session identifier and the raw response are in the release.
617 The model identifier and inference settings are not exposed by the platform
618 and are operator-declared (Anthropic Claude Sonnet 4.6).

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624 of the author.

625 **Declaration of Generative AI and AI-Assisted Technologies in the** 626 **Manuscript Preparation Process**

627 During the preparation of this work the author used OpenAI GPT-5.6
628 and Anthropic Claude (Sonnet 5.5), accessed through the OpenCode coding
629 assistant, in order to edit language, restructure text, check consistency, verify
630 citations against Crossref metadata, proof the LaTeX source, and write and
631 run the analysis scripts that produce the reported evidence. These assistants
632 are distinct from the Claude Sonnet 4.6 deployment evaluated in the study.
633 After using these tools, the author reviewed and edited the content as needed
634 and takes full responsibility for the content of the published article. The
635 figures are data visualizations drawn by scripts directly from the reported
636 evidence; no generative AI created or altered any image.

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